For Phase 1, I used the Claude Opus 5 model and found two concerning patterns throughout my conversation: it always answered questions with too much confidence and took shortcuts whenever possible. When I prompted it to generate relevant research papers, it often led me to arXiv, presumably because papers there are not peer-reviewed and therefore easier to access. When I asked Claude to quiz me, I noticed after the second paper's quiz that the correct answer was always the longest one. When I mentioned this, Claude confirmed it and also noted that all the correct answers were either (b) or (c). This was corrected for the third paper's quiz, and I even got some of the questions on that quiz wrong.

When the LLM found potential research papers, it "skimmed" them or did not read them, which would not have been a problem had it not also suggested research topics. At one point, it recommended a topic, but when I read one of the papers and asked it to quiz me, it realized that the topic had already been explored. I then asked it to generate a tutorial for the topic based on the three papers, which it did, and I then reviewed for accuracy (by reviewing the papers and checking citations) and wording (the tutorial was full of short, "cocky" statements). I corrected the tutorial, reuploaded it, and asked Claude to review it, but it realized that it mislabeled information, overexaggerated claims, and even miscited information to a paper outside the three I read (I fixed these by rereading the papers and confirming everything written).

I found the best prompting approach was making the LLM look at everything twice, whether asking it to quiz me on a paper it found or telling it to reread a tutorial it generated. On the second go-around, the LLM always caught things it skipped the first time. The best way of getting a result seems to be interrogating it repeatedly and forcing it to review. However, one thing I learned from reading the papers that the LLM seemed to miss was the context they were written in. All the papers had an abstract or background section that helped frame the problem, but the LLM skipped past those parts.

After this phase, I want to do a project in the same vein as the "AI for Health & Medical Imaging" topic, but with patents. The project would be capable of seeing similarities in patent drawings, diagrams, or text within a specific field (since it is only a semester-long project). There is a large, reliable, and free database of relevant files through the United States Patent & Trademark Office, so I believe something focused on a specific subject could be possible within the semester's 12-week timeframe.